



Adiponectin is related to markers of endothelial dysfunction and neoangiogenesis in diabetic patients

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Abstract

Purpose Adiponectin an adipokine, produced by mature adipocyte, has an important effect on several aspects of endothelial function, including leukocyte adhesion (mediated by adhesion molecules like intercellular adhesion molecule 1 (ICAM1) endothelial cell selective adhesion molecule ESAM). Recently, it has been linked to vascular endothelial growth factor (VEGF)-modulated angiogenesis. ESAM might also be involved in modulating VEGF-dependent actions. We studied relationship of adiponectin to ESAM, ICAM1, and VEGF in type 2 diabetic patients (T2DP) with or without microvascular complications.

Methods Incident T2DP referred for nephrologic evaluation were included (patients with no nephropathy or stage 1–4 nephropathy). T2DP with stage 5 chronic kidney disease (CKD) were selected from a dialysis center. Clinical, standard laboratory assessment and adiponectin, ESAM, ICAM1, and VEGF (ELISA) were recorded.

Results Eighty-seven patients were included, 15 had no CKD, 30 with stage 1 or 2 CKD, 20 with stage 3 or 4 CKD and 22 patients on dialysis. ESAM was higher in patients with CKD than in those without CKD ($p=0.02$), adiponectin, ICAM1, and VEGF were similar. Adiponectin correlated in univariate analysis to ESAM ($r=0.32$, $p=0.002$), ICAM1 ($r=0.23$, $p=0.038$), and CRP ($r=0.31$, $p=0.012$), and inversely to serum albumin ($r=-0.57$, $p<0.0001$). In predialysis patients, adiponectin also correlated to albuminuria ($r=0.54$, $p<0.0001$) and glomerular filtration rate ($r=-0.46$, $p=0.0001$). In multivariate regression ESAM ($p=0.04$), VEGF ($p=0.03$), and albumin ($p<0.0001$) are significant predictors of adiponectin. None of these cytokines were different when comparing patients with and without retinopathy.

Conclusion Adiponectin is directly linked to adhesion molecules and VEGF in T2DP.

Keywords Diabetes · Chronic kidney disease · Adiponectin · Atherogenesis · ESAM · Adhesion molecule

Introduction

Diabetes mellitus presents with an array of microvascular and macrovascular complications that are responsible for its dramatically increasing morbidity and mortality [1]. One of the main causes of these vascular complications is hyperglycemia-induced endothelial dysfunction, which is especially prominent in patients with diabetes-associated kidney disease. An important factor influencing endothelial function is adiponectin, produced by mature adipocytes. Adiponectin has been extensively studied with respect to its insulin-sensitizing effect [2], and has been incriminated in the development of metabolic conditions like type 2 diabetes and obesity [3, 4], but the deficiency of adiponectin has also been associated with certain aspects of endothelial

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function like increased oxidative stress, inflammation, and atherogenesis [5, 6].

Two further aspects of endothelial function seem to be regulated by adiponectin [7]. First, adiponectin influences endothelial adhesion and transmigration of leukocytes and macrophages, mediated by adhesion molecules like intercellular adhesion molecule 1 (ICAM 1), vascular endothelial growth factor (VCAM), and selectins. Second, adiponectin directly modulates VEGF-dependent angiogenesis in vitro [7] and may therefore be relevant in postischemic response, microvascular complications, or cancer development in diabetic patients. One more recently described adhesion molecule, the endothelial cell selective adhesion molecule (ESAM), has important function not only in cellular adhesion but also in assuring integrity of the endothelium and might be involved in modulating VEGF dependent actions.

Clinical studies describing a possible relation of adiponectin to endothelial cell adhesion molecules and VEGF are scarce and its relationship to ESAM has never been assessed.

In our study, we aim to describe a possible link of serum adiponectin to markers of endothelial dysfunction and VEGF in a cohort of diabetic patients with different stages of nephropathy.

Subjects and methods

Patients included in a previous transversal evaluation [8] were included in the present study if enough frozen serum was available. Included patients had diagnosis of type 2 diabetes for at least 6 months, without chronic kidney disease (CKD) or with stages 1–4 CKD recruited from incident patients referred for nephrology evaluation, as well as stage 5 CKD that were selected from a dialysis unit. Exclusion criteria were the presence of urinary abnormalities except albuminuria, urinary tract obstruction, polycystic disease or other known cause of CKD, recent (past 3 months) infections and any other serious chronic or acute disease requiring treatment. Medical records were analyzed to obtain history and evidence for the presence or absence of diabetic retinopathy. Blood samples were collected in the morning after 8-h fasting and maintained at 4 °C for ≤ 4 h. Standard laboratory assessment was performed right away; plasma and serum were frozen at -70 °C. Creatinine, total and high-density lipoprotein cholesterol, triglycerides (automated colorimetric enzymatic method), C-reactive protein (CRP), serum albumin and glycated hemoglobin (immunoturbidimetric method) and urinary multistick analysis (COMBUR) were performed. Low-density lipoprotein (LDL) cholesterol was calculated according to Friedewald formula [9]. Glomerular filtration rate (GFR) was estimated according to abbreviated CKD-EPI formula [10]. Urinary albumin to creatinine

ratio (UACR) was calculated as mean of three consecutive morning spot measurements. Cytokines were determined in serum by enzyme-linked immunosorbent assay (ELISA): for adiponectin (R&D System) with inter-assay variability $< 6.9\%$, intra-assay variability $< 4.7\%$, lower limit of detection 0.246 ng/ml, for ESAM (Uscn Life Science Inc) with minimum detection limit 0.118 ng/ml, intra-assay variability $< 10\%$ and inter-assay variability $< 12\%$, for ICAM 1 lower inter-assay variability < 6.7 , intra-assay variability $< 5.2\%$, with a lower detection limit of 0.049 ng/ml, for VEGF-A (Cederlane) with minimum detection limit 15.6 pg/ml, inter- and intra-assay variability $< 10\%$.

The study was approved by the ethical committee of our University and is in accordance with the ethical standards of the Declaration of Helsinki; each patient signed an informed consent.

Statistical analysis was performed using SPSS 13.0, Statistica 7.0 and Microsoft EXCEL programs. Normal distribution was tested with Kolmogorov–Smirnov test. Values are expressed as mean $\pm$ standard deviation for normally distributed variables, respectively, as median (interquartile range) for non-normally distributed variables. For comparison of two means of independent samples, *t* test or Man–Whitney test was used. For comparison of qualitative variables, χ^2 test or Fisher exact test was employed. For identifying correlation between two continuous variables, Pearson's or Spearman correlation coefficient (*r*) was assessed. This was followed by multivariate linear regression using Enter method. Statistical significance threshold was considered $\alpha = 0.05$.

Results

We included 87 patients out of which 15 had no evidence of CKD according to KDIGO criteria, 30 patients had stage 1 or 2 KDIGO CKD, 20 patients had stage 3 or 4 CKD and 22 patients had stage 5 CKD and were on dialysis. We compared patients with CKD to patients without evidence of nephropathy. The characteristics of the cohort and comparison of patients with and without CKD is depicted in Table 1. There were no significant differences in adiponectin, adhesion molecules, and VEGF in patients with and without nephropathy with the exception of ESAM which was higher CKD patients. However, as eGFR or UACR were not available in these patients, for these parameters the comparison was performed between patients without nephropathy and predialysis CKD patients. eGFR was as expected smaller in predialysis CKD patients -52.3 (8.7–85.30) versus 95.1 (92.55–99.85) ml/min, $p < 0.0001$ and UACR higher -37.94 (15.14–100.917) versus 12.37 (3.45–20.69) mg/g, $p = 0.001$ than in patients without CKD.

Adiponectin correlated significantly in univariate analysis to ESAM ($r = 0.32$, $p = 0.002$), and ICAM 1 ($r = 0.23$,

Table 1 General characteristics of the cohort

Parameter	All patients (<i>n</i> = 87)	Without nephropathy (<i>n</i> = 15)	With nephropathy (<i>n</i> = 72)	<i>p</i>
Age (years)	62.83 ± 9.92	58.20 ± 11.48	63.79 ± 9.37	0.16
Gender <i>n</i> (%)	51 (58.62)	9 (60.00)	42 (58.33)	0.90
Diabetes duration (years)	6.00 (4.00–15.00)	5.50 (4.00–10.00)	7.00 (4.00–15.00)	0.20
Adiponectin (µg/ml)	6.69 (3.57–18.79)	5.10 (3.22–6.05)	7.38 (3.70–19.10)	0.15
ESAM (ng/ml)	0.70 (0.20–8.90)	0.20 (0.10–0.80)	1.00 (0.25–9.30)	0.02
VEGF (pg/ml)	73.30 (56.00–110.00)	79.90 (58.05–120.70)	72.30 (55.08–109)	0.74
ICAM 1 (ng/ml)	225 (188.00–276.50)	220 (185.00–309.00)	226 (191.00–267.00)	0.97
Hb A1C (%)	7.45 ± 1.30	8.11 ± 1.57	7.31 ± 1.20	0.03
Serum albumin (g/dl)	4.50 (4.07–4.70)	4.50 (4.45–4.90)	4.40 (3.95–4.70)	0.06
LDL cholesterol (mg/dl)	103.39 ± 38.19	97.76 ± 30.87	105.11 ± 40.29	0.51
HDL cholesterol (mg/dl)	41.74 ± 13.30	43.27 ± 7.45	41.39 ± 14.32	0.62
Triglycerides (mg/dl)	159.00 (123.00–204.00)	146 (112.20–208.50)	159 (126.50–203.50)	0.50
CRP (mg/dl)	0.60 (0.20–1.10)	0.4 (0.10–0.70)	0.88 (0.60–1.87)	0.29

Values are expressed as mean ± SD for normally distributed variables and median (25th–75th percentiles) for the abnormally distributes ones
ESAM endothelial cell selective adhesion molecule, *VEGF* vascular endothelial growth factor, *ICAM 1* intercellular adhesion molecule 1, *Hb A1C* glycated hemoglobin, *LDL* low-density lipoprotein, *HDL* high-density lipoprotein, *CRP* C reactive protein

$p = 0.038$). Adiponectin also correlated inversely to serum albumin ($r = -0.57$, $p < 0.0001$), eGFR ($r = -0.46$, $p = 0.0001$), and directly to UACR ($r = 0.54$, $p < 0.0001$) and CRP ($r = 0.31$, $p = 0.012$). ESAM as significantly related to VEGF ($r = 0.25$, $p = 0.020$), inversely to eGFR ($r = -0.62$, $p < 0.0001$), but not to UACR. ICAM 1 and VEGF did not seem to have a significant relationship to either kidney function or albuminuria.

In multivariate regression—enter method, with adiponectin as a dependent variable and independent variables those significantly correlated to adiponectin in univariate analysis and available in all patients, ESAM ($p = 0.04$), VEGF ($p = 0.03$), and albumin ($p < 0.0001$) remain significant predictors of adiponectin (Table 2).

In 72 patients, information about presence of diabetic retinopathy (DR) was available. Patients with and without DR were compared: there were no significant differences in what concerns adiponectin, adhesion molecules or VEGF.

Comparison is presented in Table 3. In 50 non-HD patients, eGFR and UACR could be calculated; information about DR was available. There were no significant differences when comparing patients without DR to those with DR in either eGFR: 64.22 (10.6–92.12) versus 52.3 (6.85–85.55) ml/min, $p = 0.26$ or UACR: 20.83 (9.26–74.72) versus 21.63 (7.06–49.58) mg/g, $p = 0.69$.

Discussion

Type 2 diabetes is recognized as a state of adiponectin deficiency. However, in patients with diabetic nephropathy adiponectin generally increases with decrease of renal function, fact confirmed in our study by the strong inverse relationship of adiponectin to eGFR. However, this increase cannot be entirely explained by reduced renal clearance, as adiponectin increases early in kidney disease, even when GFR is normal

Table 2 Multivariate regression analysis with adiponectin as dependent variable

	Unstandardized Coefficients		Standardized Coefficients	<i>p</i>	95% confidence interval for B	
	B	Std. error			Lower bound	Upper bound
(Constant)	51.742	10.781			30.234	73.250
ESAM	0.117	0.057	0.232	0.044	0.003	0.232
ICAM 1	0.010	0.009	0.109	0.259	− 0.007	0.027
VEGF	− 0.040	0.017	− 0.231	0.026	− 0.074	− 0.005
Albumin	− 9.368	2.265	− 0.457	< 0.001	− 13.886	− 4.85
CRP	0.219	0.436	0.049	0.617	− 0.651	1.089

Std. error standard error, *ESAM* endothelial cell selective adhesion molecule, *ICAM 1* intercellular adhesion molecule 1, *VEGF* vascular endothelial growth factor, *CRP* C reactive protein

Table 3 Characteristics of the patients with and without retinopathy

Parameter	Without retinopathy (<i>n</i> = 45)	With retinopathy (<i>n</i> = 27)	<i>p</i>
Age (years)	62.31 ± 9.01	65.21 ± 9.29	0.04
Gender <i>n</i> (%)	26 (57.77)	15 (55.55)	0.84
Diabetes duration (years)	6.00 (4.00–10.00)	12.00 (6.00–18.00)	0.07
Adiponectin (µg/ml)	7.39 (3.88–19.59)	7.37 (4.20–16.87)	0.83
ESAM (ng/ml)	0.70 (0.30–9.10)	2.50 (0.25–18.30)	0.31
VEGF (pg/ml)	65.90 (55.60–108.30)	73.30 (51.30–114.30)	0.88
ICAM 1 (ng/ml)	230 (204.00–268.00)	223 (172.00–278.00)	0.62
Hb A1C (%)	7.30 ± 1.24	7.50 ± 1.15	0.49
Serum albumin (g/dl)	4.40 (4.03–4.70)	4.30 (3.80–4.70)	0.81
LDL cholesterol (mg/dl)	103.46 ± 37.93	102.58 ± 36.39	0.93
HDL cholesterol (mg/dl)	40.42 ± 12.64	44.32 ± 16.25	0.26
Triglycerides (mg/dl)	166 (129.00–213.00)	138 (89.50–17.50)	0.03
CRP (mg/dl)	0.67 (0.20–1.10)	0.60 (0.30–1.80)	0.81

Values are expressed as mean ± SD for normally distributed variables and median (25th–75th percentiles) for the abnormally distributed ones, or *n* (percentage)

ESAM endothelial cell selective adhesion molecule, *VEGF* vascular endothelial growth factor, *ICAM 1* intercellular adhesion molecule 1, *Hb A1C* glycated hemoglobin, *LDL* low-density lipoprotein, *HDL* high-density lipoprotein, *CRP* C reactive protein

and is closely linked to increased protein excretion [11–13], relationship which is demonstrable in our study also. The direct correlation of adiponectin with UACR is the probable explanation for the inverse relationship of serum albumin to adiponectin found in multivariate regression, given that increase UACR and decrease in serum albumin are linked pathogenically and statically ($r = -0.6$, $p < 0.0001$). One possible explanation for the unexpected high adiponectin–high risk state of CKD is as an adaptive response resulting in increase of adiponectin triggered by inflammation and endothelial dysfunction and aimed at mitigating these pathological processes [14, 15]. This adaptive increase in adiponectin would have beneficial effects on several aspects of endothelial function, like oxidative stress [5], inflammation, leukocyte adhesion and transmigration, platelet adherence and atherogenesis. This epidemiologic conundrum could explain conflicting result reported in studies regarding the in vivo association of adiponectin to circulating adhesion molecules like VCAM or ICAM 1. In agreement to some, but not all of the data in the literature [16–22] adiponectin closely related to ICAM in our study. However, in experimental settings, adiponectin deficiency has been consistently demonstrated to activate cell adhesion in animal models, and replacement of adiponectin to counteract this effect [23, 24]. Differential effects of full length and globular adiponectin as well as local generation of globular adiponectin in response to inflammation might be an important contributor to the often conflicting results reported in clinical studies [7]. The most important finding in our study was the strong relationship of adiponectin to ESAM, in univariate and multivariate analyses. ESAM is a particular adhesion molecule, important not only in leukocyte adhesion but also in assuring

integrity of endothelium—as it binds together endothelial cells at the level of the tight junction—in mediating platelet adherence and angiogenesis and regulating endothelial permeability. ESAM has a soluble fraction and higher levels of serum ESAM have been associated to atherosclerosis and its complications [25, 26]. Our group has previously described ESAM in diabetic patients with and without nephropathy and its direct relationship to VEGF in patients with diabetes associated kidney disease [8]. Both adiponectin and ESAM are involved in angiogenesis, which is tightly regulated by VEGF. Neoangiogenesis is an essential component in microvascular disease, helps mitigate post-ischemic injury, and is important in cancer development and metastasis. ESAM has been previously involved in angiogenesis [25, 27, 28]. There is preliminary evidence that ESAM might be needed for VEGF-induced signal transduction and facilitating VEGF-induced effects on the endothelium [29]. On the other hand, adiponectin has been reported to promote angiogenesis both in vitro and in vivo via adenosine monophosphate-activated protein kinase—endothelial nitric-oxide synthase (AMPK-eNOS) pathway [30–33]. Recently, adiponectin has been demonstrated to induce VEGF-A expression and subsequently promote angiogenesis and tumor growth in human chondrosarcoma cells through the activation of AdipoR1/R2 receptor, phosphatidylinositol 3-kinases (PI3K), protein kinase B (Akt), mammalian target of rapamycin (mTOR), and hypoxia-inducible factor 1- α (HIF-1 α) signaling pathways. Furthermore, knock-down of adiponectin decreased adiponectin-induced angiogenesis in vitro [34].

Thus, the relation of adiponectin to ESAM and VEGF, together with direct correlation of ESAM to VEGF found in our study and in line with the above-mentioned experimental

data raises the question of a possible pathogenic interaction between adiponectin and ESAM/VEGF-mediated angiogenesis. However, the positive relation of adiponectin to ESAM might be also explain through the reactive increase in adiponectin in relation to inflammation, endothelial dysfunction, leukocyte adhesion, and atherosclerosis in diabetes.

This direct relationship between ESAM and adiponectin found in our study has not, to our knowledge been previously reported. The only other association of adiponectin to ESAM comes from acute myocardial infarction biomarker study in which the risk for development of acute kidney injury in ST-elevation myocardial infarction patients undergoing percutaneous coronary intervention is associated with higher levels of both adiponectin and ESAM at baseline [35].

Since neoangiogenesis and endothelial proliferation are important pathogenic events in diabetic retinopathy and anti-VEGF therapy is a mainstay of treatment [36] we compared patients with/without diabetic retinopathy.

In our study, we did not find a clinically significant difference between adiponectin, VEGF or soluble fraction of adhesion molecules. We need to emphasize that presence of retinopathy was recorded according to available antecedent findings in the medical documents of the patients; however, these records were neither contemporary with inclusion of patients, nor mandatory (data missing in 15 patients) and they were not verified upon inclusion, which warrants confirmation by further studies.

One of the shortcomings of our study is the relatively limited number of patients in our study, as is the wide array of CKD stage in our patients, larger studies including more homogenous population are needed to confirm our results.

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Compliance with ethical standards

Conflict of interest All authors declared that they have no conflicts of interest.

Ethical approval The study was in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki declaration and its later amendments; all included patients signed an informed consent.

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